

Rationality and growth of stabilized permutation-weight series

Finite spectra, integer recurrences, and an explicit two-descent formula

Research report prepared for Vladimir Reshetnikov

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Abstract

The permutation-weight q -Eulerian polynomials have stabilized leading coefficients $a_{d,k}$ when the number of descents d is fixed. Earlier work proved stabilization and investigated an initial agreement with the partition triangle OEIS A256193. We pursue the complete-series problem: can the entire series $W_d(t) = \sum_{k \geq 0} a_{d,k} t^k$, including the coefficients beyond that agreement, be described by a finite exact formula?

Starting from the published polynomial recurrence, we prove that every W_d is rational. The key is a finite exponential spectrum indexed by partitions of integers at most $d + 1$. We give an explicit universal denominator, an exact construction of the numerator, and a practical symbolic algorithm. We also prove that the radius of convergence is exactly $(d + 1)^{-1/d}$, with a periodic leading asymptotic term. For $d = 2$ we obtain closed formulas involving only 3^m , 2^m , and Fibonacci numbers. In contrast, the fixed- d diagonal series of A256193 is nonrational and has radius 1. Exact computations for $1 \leq d \leq 6$ are independently checked against the original recurrence through deficit 40; the archive contains all code, certificates, rational functions, and coefficient data.

Scope and status. The stabilization theorem itself is known, and the earlier near-diagonal partition correspondence is not claimed as new. The problem selected here is a precise generating-function continuation of the request for further coefficients in Huang–Tang: determine each *whole* stabilized series by finite algebraic data. The rationality theorem, denominator construction, and growth analysis below were derived in this investigation and were not found in the three primary papers reviewed. This is a proved research result within the report, not an exhaustive certification of novelty or a claim of peer review. The English proofs do not depend on numerical fitting or on an unproved conjecture.

Companion files: `code/derive.py`, `code/verify.py`, `data/rational_series.json`.

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The principal conclusions at a glance. For every fixed integer $d \geq 1$,

$$W_d(t) \in \mathbb{Q}(t) \cap \mathbb{Z}_{\geq 0}[[t]], \quad \text{rad}(W_d) = (d+1)^{-1/d}, \quad \limsup_{k \rightarrow \infty} a_{d,k}^{1/k} = (d+1)^{1/d}.$$

For the two-descent sequence, with $F_0 = 0$, $F_1 = 1$,

$$\begin{aligned} a_{2,2m} &= 45 \cdot 3^m + 8 \cdot 2^m - F_{2m+10} + m + 3, \\ a_{2,2m+1} &= 81 \cdot 3^m + 8 \cdot 2^m - F_{2m+11} + m + 4. \end{aligned}$$

These identities hold for every $m \geq 0$, not merely over a tested range.

1 The selected problem and its provenance

1.1 A specific family, not an arbitrary q -Eulerian convention

There are several inequivalent q -analogues of Eulerian polynomials. Here the family is the permutation-weight family introduced by Dugan, Glennon, Gunnells, and Steingrímsson [1]. Its terms enumerate permutations by descents and a recursively defined weight arising from maxmin trees. To remove any ambiguity about the statistic or normalization, we take the recurrence established by Agrawal, Choi, and Sun [2, Theorem 4.1] as the exact algebraic specification:

$$\begin{aligned} E_0(x, q) &= 1, \\ E_n(x, q) &= E_{n-1}(qx, q) \\ &\quad + x \sum_{i=1}^{n-1} \binom{n-1}{i} E_i(x, q) E_{n-i-1}(qx, q) \quad (n \geq 1). \end{aligned} \tag{1.1}$$

Write

$$A_{n,d}(q) = [x^d] E_n(x, q), \quad D(n, d) = d(n - d - 1). \tag{1.2}$$

For $n \geq d + 1$, the degree of $A_{n,d}$ is $D(n, d)$ and its leading coefficient is 1. Reversing the polynomial gives

$$R_{n,d}(t) = t^{D(n,d)} A_{n,d}(t^{-1}). \tag{1.3}$$

The coefficient of t^k counts weight deficit k below the largest weight available at length n and descent count d .

The coefficients stabilize as n increases. In fact, the coefficient of t^k is fixed from $n = d + k + 1$ onward. Define

$$a_{d,k} = [t^k] R_{d+k+1,d}(t), \quad W_d(t) = \sum_{k \geq 0} a_{d,k} t^k \quad (d \geq 1). \tag{1.4}$$

This stabilization is due to Agrawal–Choi–Sun [2]. We reproduce a short proof in [section 2](#), both to make the argument self-contained from (1.1) and to justify the independent verification algorithm.

1.2 What was already known, and what is investigated here

Dugan et al. observed stabilization and a partial agreement with the partition triangle A256193 [1, Section 6.12]. Agrawal et al. proved stabilization and proposed a recurrence relating certain near-diagonal coefficients [2, Section 6]. Huang and Tang subsequently gave a maxmin-tree treatment of the partition connection; their final discussion asks for ways to reach further coefficients when more complicated tree structures enter [3, Section 5]. Thus it would be incorrect to present stabilization or the original partition correspondence as an unresolved conjecture in this report.

The question pursued here is a more specific continuation:

Problem 1.1. For fixed d , determine whether the *entire* stabilized sequence $(a_{d,k})_{k \geq 0}$ satisfies a finite constant-coefficient linear recurrence. If it does, construct one exactly, and determine its exponential growth.

This formulation is ours; it is not attributed to the earlier authors as a verbatim named conjecture. It asks for information substantially stronger than computing a finite prefix from (1.1). The answer will be affirmative for every d .

2 Leading terms and stabilization

Taking the coefficient of x^d in (1.1) gives

$$A_{n,d} = q^d A_{n-1,d} + \sum_{i=1}^{n-1} \binom{n-1}{i} \sum_{a+b=d-1} q^b A_{i,a} A_{n-i-1,b}. \quad (2.1)$$

All omitted arguments are q . We use $A_{0,0} = 1$, $A_{0,d} = 0$ for $d > 0$, and $A_{n,d} = 0$ outside $0 \leq d \leq n-1$ when $n \geq 1$.

Lemma 2.1. *For $n \geq d+1$ and $d \geq 1$, $A_{n,d}$ has nonnegative integer coefficients, degree $D(n,d)$, and leading coefficient 1. Also $A_{n,0} = 1$ for every $n \geq 0$.*

Proof. Positivity and the assertion for $d = 0$ follow immediately by induction. The support bound $d \leq n-1$ also follows from (2.1). At $n = d+1$, the only nonzero convolution term is $i = d$, $a = d-1$, and an empty right factor, so $A_{d+1,d} = 1$.

For the degree comparison, let $m = n - i - 1$. First suppose $m > 0$. For a nonzero term write

$$i = a + 1 + u, \quad m = b + 1 + v, \quad u, v \geq 0, \quad a + b = d - 1.$$

The difference between $D(n,d)$ and the maximal degree of $q^b A_{i,a} A_{m,b}$ is

$$\Delta = a + 1 + (b + 1)u + (a + 1)v. \quad (2.2)$$

This is positive. If $m = 0$, then $b = 0$, $a = d - 1$, and the same difference is

$$\Delta = n - d - 1. \quad (2.3)$$

For $n > d+1$ this is again positive. On the other hand, $q^d A_{n-1,d}$ has degree exactly $D(n,d)$ and leading coefficient 1. It is therefore the unique contribution to the leading term. $\square$

Proposition 2.2 (Known stabilization, reproved). *For $n \geq d+2$,*

$$R_{n,d}(t) - R_{n-1,d}(t) \in t^{n-d-1} \mathbb{Z}_{\geq 0}[t]. \quad (2.4)$$

Consequently $[t^k] R_{n,d}$ is independent of n for $n \geq d+k+1$.

Proof. After reversal, the first term of (2.1) is exactly $R_{n-1,d}$. A nonempty-right convolution term becomes a product of reversed polynomials multiplied by t^Δ , where (2.2) gives

$$\Delta - (n - d - 1) = (a + 1 + (b + 1)u + (a + 1)v) - (1 + u + v) = a + bu + av \geq 0.$$

An empty-right term has exactly the shift in (2.3). All coefficients are nonnegative. Thus the increment at length n vanishes in degrees below $n - d - 1$. Once $n = d + k + 1$ has been reached, every subsequent increment starts in degree at least $k + 1$. $\square$

The limit $R_{n,d} \rightarrow W_d$ is a *formal* power-series limit: each fixed coefficient eventually stops changing. No numerical value of t , analytic convergence, or interchange of infinite analytic sums is needed at this stage.

3 The finite exponential spectrum

3.1 The differential hierarchy

Introduce exponential generating functions in the length variable:

$$\mathcal{F}(z, x, q) = \sum_{n \geq 0} E_n(x, q) \frac{z^n}{n!}, \quad f_d(z, q) = [x^d] \mathcal{F}(z, x, q). \quad (3.1)$$

All calculations in this section are over the field $K = \mathbb{Q}(q)$. The binomial convolution in (1.1) becomes multiplication:

$$\partial_z \mathcal{F}(z, x, q) = (1 + x(\mathcal{F}(z, x, q) - 1)) \mathcal{F}(z, qx, q). \quad (3.2)$$

Hence $f_0 = e^z$, while, for $d \geq 1$,

$$\partial_z f_d = q^d f_d + g_d, \quad f_d(0, q) = 0, \quad (3.3)$$

where

$$g_d(z, q) = \sum_{a+b=d-1} q^b f_a(z, q) f_b(z, q) - q^{d-1} f_{d-1}(z, q). \quad (3.4)$$

The subtracted term is essential: it removes the forbidden empty left factor in the original recurrence. Omitting it produces a different sequence.

3.2 A partition-indexed set of possible frequencies

Definition 3.1. For $d \geq 0$, let

$$\Lambda_d = \left\{ \lambda_c(q) = \sum_{j=0}^d c_j q^j : c_j \in \mathbb{Z}_{\geq 0}, \quad 1 \leq \sum_{j=0}^d (j+1)c_j \leq d+1 \right\}. \quad (3.5)$$

The integer $\sum (j+1)c_j$ is the cost of the frequency. Write $M_d = |\Lambda_d|$.

A vector $\mathbf{c}$ of cost r is exactly a partition of r with c_j parts of size $j+1$. Distinct vectors give distinct polynomials in q . Therefore

$$M_d = \sum_{r=1}^{d+1} p(r), \quad (3.6)$$

where $p(r)$ is the ordinary partition number. The first values are $M_0, M_1, \dots, M_6 = 1, 3, 6, 11, 18, 29, 44$.

For example,

$$\begin{aligned} \Lambda_1 &= \{1, 2, q\}, \\ \Lambda_2 &= \{1, 2, 3, q, q+1, q^2\}, \\ \Lambda_3 &= \{1, 2, 3, 4, q, q+1, q+2, 2q, q^2, q^2+1, q^3\}. \end{aligned}$$

The only frequency in Λ_d of degree d is q^d . Every other frequency has degree at most $d-1$.

Theorem 3.2 (Finite-spectrum theorem). *For every $d \geq 0$, there are rational functions $c_{d,\lambda}(q) \in \mathbb{Q}(q)$ such that*

$$f_d(z, q) = \sum_{\lambda \in \Lambda_d} c_{d,\lambda}(q) e^{\lambda(q)z}. \quad (3.7)$$

In particular,

$$A_{n,d}(q) = \sum_{\lambda \in \Lambda_d} c_{d,\lambda}(q) \lambda(q)^n \quad (n \geq 0). \quad (3.8)$$

Some allowed coefficients may be zero; the theorem does not assert that every allowed frequency occurs for every d .

Proof. For $d = 0$, $f_0 = e^z$ and $\Lambda_0 = \{1\}$. Suppose the assertion holds at every smaller descent count. If $a + b = d - 1$, a frequency from $f_a f_b$ is a sum $\lambda + \mu$ of costs at most $(a + 1) + (b + 1) = d + 1$. It belongs to Λ_d . Frequencies from f_{d-1} also belong to Λ_d . None of these has degree d : both factors in a product have descent count at most $d - 1$.

Write $g_d = \sum_{\lambda} h_{d,\lambda} e^{\lambda z}$ after collecting equal frequencies. Solving (3.3) gives the finite expression

$$f_d(z, q) = \sum_{\lambda} \frac{h_{d,\lambda}(q)}{\lambda(q) - q^d} (e^{\lambda(q)z} - e^{q^d z}). \quad (3.9)$$

Every denominator is nonzero in $\mathbb{Q}(q)$, because $\deg \lambda < d$. The only new frequency is q^d , whose cost is $d + 1$. This proves (3.7); coefficient extraction in z gives (3.8). $\square$

Distinct exponentials over K are linearly independent: comparing the first M_d derivatives at $z = 0$ gives a Vandermonde matrix with nonzero determinant. Thus the collected coefficients in (3.7) are unique. The absence of resonance in (3.9) is an identity in the indeterminate q , not an assertion that particular numerical specializations of q can never produce coincident frequencies.

4 Rationality and a universal integer denominator

Let

$$C_d(q) = c_{d,q^d}(q). \quad (4.1)$$

The coefficient C_d is distinguished by degree, not by ordering frequencies at a particular real value of q .

Theorem 4.1 (Rationality of the complete stabilized series). *For every integer $d \geq 1$,*

$$\boxed{W_d(t) = t^{-d(d+1)} C_d(t^{-1})}. \quad (4.2)$$

Consequently $W_d \in \mathbb{Q}(t) \cap \mathbb{Z}_{\geq 0}[[t]]$.

Proof. Substitute (3.8) into (1.3):

$$R_{n,d}(t) = t^{-d(d+1)} \sum_{\lambda \in \Lambda_d} c_{d,\lambda}(t^{-1}) (t^d \lambda(t^{-1}))^n. \quad (4.3)$$

For $\lambda = q^d$, the parenthesized factor is 1. For every other frequency it is a polynomial divisible by t , since $\deg \lambda \leq d - 1$. Each rational amplitude, viewed in $\mathbb{Q}((t))$, has a finite Laurent principal part. Multiplication by the n th power of a polynomial of positive t -adic order therefore tends coefficientwise to zero. There are finitely many frequencies, so only the q^d term survives. By theorem 2.2, the limit on the left is W_d . Its integrality, nonnegativity, and regularity at $t = 0$ were already established directly from the recurrence. $\square$

4.1 A denominator requiring no guesswork

Define

$$\mathcal{D}_d(t) = \prod_{\lambda \in \Lambda_d \setminus \{q^d\}} (1 - t^d \lambda(t^{-1})). \quad (4.4)$$

Every factor is an integer polynomial with constant coefficient 1.

Theorem 4.2 (Explicit denominator and finite numerator). *For every $d \geq 1$, there is a polynomial $\mathcal{P}_d(t) \in \mathbb{Z}[t]$ with*

$$W_d(t) = \frac{\mathcal{P}_d(t)}{\mathcal{D}_d(t)}. \quad (4.5)$$

The numerator can be constructed from the finitely many polynomials $A_{0,d}, \dots, A_{M_d-1,d}$ as follows. Put

$$B_d(u, q) = \prod_{\lambda \in \Lambda_d \setminus \{q^d\}} (u - \lambda(q)) = \sum_{j=0}^{M_d-1} b_{d,j}(q) u^j, \quad (4.6)$$

then set

$$\begin{aligned} N_d(q) &= \sum_{j=0}^{M_d-1} b_{d,j}(q) A_{j,d}(q), \\ \mathcal{P}_d(t) &= t^{d(M_d-1)-d(d+1)} N_d(t^{-1}). \end{aligned} \quad (4.7)$$

Although the last expression is initially written as a Laurent polynomial, all its negative powers cancel.

Proof. Apply the polynomial B_d to the power-sum representation (3.8) at $n = 0$. It annihilates every frequency except q^d :

$$N_d(q) = C_d(q) B_d(q^d, q).$$

Thus

$$C_d(q) = \frac{N_d(q)}{\prod_{\lambda \neq q^d} (q^d - \lambda(q))}. \quad (4.8)$$

Substitution into (4.2) gives (4.5) and (4.7). All $b_{d,j}$ and $A_{j,d}$ are integer polynomials, so the expression for $\mathcal{P}_d$ is a Laurent polynomial with integer coefficients. But $\mathcal{P}_d = \mathcal{D}_d W_d$, with $\mathcal{D}_d \in \mathbb{Z}[t]$ and $W_d \in \mathbb{Z}[[t]]$. It has no negative powers, proving the claim. $\square$

Corollary 4.3 (An explicit constant-coefficient recurrence). *Write $\mathcal{D}_d(t) = 1 + \delta_1 t + \dots + \delta_L t^L$. Then, for every $k > \deg \mathcal{P}_d$,*

$$a_{d,k} + \sum_{j=1}^L \delta_j a_{d,k-j} = 0, \quad (4.9)$$

where coefficients with negative indices are set equal to zero. In particular, a conventional recurrence in the preceding L terms holds once $k \geq \max(L, \deg \mathcal{P}_d + 1)$.

Proof. Compare coefficients in $\mathcal{D}_d W_d = \mathcal{P}_d$. $\square$

This is a constructive answer to theorem 1.1. The order bound $L \leq d(M_d - 1)$ is finite and explicit. It need not be minimal: cancellation already occurs at $d = 4$. For computation, the sparse differential construction (3.9) is considerably more economical than forming every $A_{j,d}$ required by (4.7). The latter formula is especially useful as a proof of the denominator bound and as a conceptually independent construction.

5 Explicit results for small descent counts

5.1 One descent

At $d = 1$, the differential equation is $f_1' = qf_1 + e^{2z} - e^z$. Hence

$$f_1(z, q) = \frac{e^{2z} - e^{qz}}{2 - q} - \frac{e^z - e^{qz}}{1 - q}, \quad C_1(q) = \frac{1}{(q-1)(q-2)}.$$

Therefore

$$W_1(t) = \frac{1}{(1-t)(1-2t)}, \quad a_{1,k} = 2^{k+1} - 1. \quad (5.1)$$

This elementary case also checks the reversal exponent in (4.2).

5.2 Two descents: the complete sequence in closed form

The next forcing term is

$$g_2(z, q) = ((q+1)e^z - q)f_1(z, q).$$

For a fully explicit extraction, write

$$f_1 = \frac{e^z}{q-1} - \frac{e^{2z}}{q-2} + \frac{e^{qz}}{(q-1)(q-2)}.$$

The five forcing frequencies are $1, 2, 3, q, q+1$. Their amplitudes are, in that order,

$$-\frac{q}{q-1}, \quad \frac{q+1}{q-1} + \frac{q}{q-2}, \quad -\frac{q+1}{q-2}, \quad -\frac{q}{(q-1)(q-2)}, \quad \frac{q+1}{(q-1)(q-2)}.$$

Summing each amplitude divided by $q^2 - \lambda$, as prescribed by (3.9), and simplifying gives

$$C_2(q) = \frac{q^3 + 2q^2 - 3q - 2}{(q-1)^2(q+1)(q^2-2)(q^2-3)(q^2-q-1)}.$$

Consequently

$$W_2(t) = \frac{1 + 2t - 3t^2 - 2t^3}{(1-t)^2(1+t)(1-2t^2)(1-3t^2)(1-t-t^2)}. \quad (5.2)$$

No interpolation of initial sequence values is used to obtain this expression: it is the exact result of solving the two differential levels.

A useful partial-fraction identity is

$$W_2(t) = \frac{8(1+t)}{1-2t^2} + \frac{9(5+9t)}{1-3t^2} - \frac{55+34t}{1-t-t^2} - \frac{1}{4(1+t)} + \frac{11}{4(1-t)} + \frac{1}{2(1-t)^2}. \quad (5.3)$$

It can be checked simply by multiplication by the denominator in (5.2).

Theorem 5.1 (Exact two-descent formula). *Let $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$. For every $m \geq 0$,*

$$a_{2,2m} = 45 \cdot 3^m + 8 \cdot 2^m - F_{2m+10} + m + 3, \quad (5.4)$$

$$a_{2,2m+1} = 81 \cdot 3^m + 8 \cdot 2^m - F_{2m+11} + m + 4. \quad (5.5)$$

Proof. The first two terms of (5.3) contribute $8 \cdot 2^m$ in both parities, and respectively $45 \cdot 3^m$ and $81 \cdot 3^m$. The coefficient of the third term is $-55F_{k+1} - 34F_k = -F_{k+10}$, using the Fibonacci addition identity. The last three terms contribute

$$-\frac{(-1)^k}{4} + \frac{11}{4} + \frac{k+1}{2},$$

which is $m+3$ for $k=2m$ and $m+4$ for $k=2m+1$. $\square$

The expanded denominator is

$$Q_2(t) = 1 - 2t - 6t^2 + 13t^3 + 11t^4 - 28t^5 - 6t^6 + 23t^7 - 6t^9.$$

Thus the ordinary recurrence of order nine is

$$\begin{aligned} a_{2,k} = & 2a_{2,k-1} + 6a_{2,k-2} - 13a_{2,k-3} - 11a_{2,k-4} \\ & + 28a_{2,k-5} + 6a_{2,k-6} - 23a_{2,k-7} + 6a_{2,k-9}, \quad k \geq 9. \end{aligned} \quad (5.6)$$

Its initial values are

$$(a_{2,0}, \dots, a_{2,8}) = (1, 4, 11, 31, 65, 157, 298, 661, 1196).$$

The order is minimal. Indeed, the numerator in (5.2) is nonzero at $t = 1, -1$; modulo $1 - 2t^2$, $1 - 3t^2$, and $1 - t - t^2$ its remainders are, respectively, $t - \frac{1}{2}$, $\frac{4}{3}t$, and t . None shares a zero with its corresponding quadratic. Therefore numerator and denominator are coprime, and no smaller eventual constant-coefficient recurrence can have this rational generating function.

An immediate asymptotic consequence is

$$a_{2,2m} \sim 45 \cdot 3^m, \quad a_{2,2m+1} \sim 81 \cdot 3^m. \quad (5.7)$$

The Fibonacci error is smaller because $\varphi^2 = (3 + \sqrt{5})/2 < 3$. In particular, $a_{2,k}^{1/k} \rightarrow \sqrt{3}$, but consecutive-term ratios do *not* converge:

$$\frac{a_{2,2m+1}}{a_{2,2m}} \rightarrow \frac{9}{5}, \quad \frac{a_{2,2m+2}}{a_{2,2m+1}} \rightarrow \frac{5}{3}.$$

A single-ratio numerical extrapolation would obscure this parity effect.

5.3 Three descents and beyond

For $d = 3$ no cancellation of the universal denominator occurs. One obtains $W_3 = P_3/Q_3$, where

$$\begin{aligned} Q_3(t) = & (1-t)(1-t^2)(1-2t^2)(1-t^3)(1-2t^3)(1-3t^3)(1-4t^3) \\ & \cdot (1-t^2-t^3)(1-t^2-2t^3)(1-t-t^3), \end{aligned} \quad (5.8)$$

$$\begin{aligned} P_3(t) = & 1 + 3t + 2t^2 - 15t^3 - 23t^4 - 17t^5 + 71t^6 + 74t^7 \\ & + 23t^8 - 117t^9 - 86t^{10} - 26t^{11} + 54t^{12} + 44t^{13} + 24t^{14}. \end{aligned} \quad (5.9)$$

The resulting recurrence has order 26. Large expanded polynomials become less readable at higher d , so their complete coefficient arrays are supplied in the companion JSON rather than typeset as several pages of integers.

The exact canceled factors for the last three rows are

$$\begin{aligned} \mathcal{D}_4/Q_4 = \mathcal{D}_5/Q_5 = & (1+t)(1+t^2), \\ \mathcal{D}_6/Q_6 = & (1+t)^4(1+t^2)(1-t+t^2)(1+t+t^2). \end{aligned}$$

These cancellations are why the universal denominator should not be advertised as the minimal denominator in general.

Table 1: Exact symbolic results. Q_d is the reduced denominator with $Q_d(0) = 1$; P_d is its coprime numerator.

d	M_d	$\deg \mathcal{D}_d$	$\deg Q_d$	$\deg P_d$
1	3	2	2	0
2	6	9	9	3
3	11	26	26	14
4	18	60	57	37
5	29	124	121	91
6	44	233	223	181

6 The exact radius of convergence

Rationality gives a finite set of possible poles. We next locate the nearest ones and prove that the positive real candidate does not cancel. Put

$$r_d = (d+1)^{1/d}, \quad \rho_d = r_d^{-1}. \quad (6.1)$$

Lemma 6.1 (Spectral separation at the first candidate pole). *For $\lambda \in \Lambda_d \setminus \{q^d\}$,*

$$\lambda(r_d) \leq d+1.$$

Equality holds only for the constant frequency $\lambda = d+1$. Consequently every factor of $\mathcal{D}_d(t)$ is nonzero for $|t| < \rho_d$. On $|t| = \rho_d$, the only possible zeros arise from $1 - (d+1)t^d$.

Proof. The function $\log(1+x)/x$ is strictly decreasing for $x > 0$. Indeed, its derivative has the sign of $x/(1+x) - \log(1+x) < 0$; the latter inequality follows by integrating $1/(1+u) > 1/(1+x)$ on $0 < u < x$. Thus, for $1 \leq j < d$,

$$r_d^j = (d+1)^{j/d} < j+1.$$

For $j = 0$, equality $r_d^0 = 1$ holds. A frequency other than q^d has $c_d = 0$, hence

$$\lambda(r_d) = \sum_{j < d} c_j r_d^j \leq \sum_{j < d} (j+1) c_j \leq d+1.$$

Equality requires that only c_0 be nonzero and $c_0 = d+1$.

For complex t with $|t| \leq \rho_d$, the triangle inequality gives

$$|t^d \lambda(t^{-1})| \leq \sum_{j < d} c_j |t|^{d-j} \leq r_d^{-d} \lambda(r_d) \leq 1.$$

The inequality is strict in the open disk, and also on its boundary unless $\lambda = d+1$. This proves the pole assertion. $\square$

6.1 Why the positive pole survives

Let $h_j(q)$ be the coefficient of $e^{(j+1)z}$ in $f_j(z, q)$. The finite-spectrum construction gives $h_0 = 1$ and

$$h_j(q) = \frac{H_j(q)}{j+1-q^j}, \quad H_j(q) = \sum_{a+b=j-1} q^b h_a(q) h_b(q) \quad (j \geq 1). \quad (6.2)$$

To see this, a product $f_a f_b$ has constant frequency $j+1$ only when its factors use their largest constant frequencies $a+1$ and $b+1$. The subtracted term in (3.4) has no such frequency.

Lemma 6.2 (Noncancellation). $C_d(q)$ has a simple pole at $q = r_d$, with residue

$$\text{Res}_{q=r_d} C_d(q) = \frac{H_d(r_d)}{d r_d^{d-1}} > 0. \quad (6.3)$$

Equivalently, the rational function

$$L_d(t) = (1 - (d+1)t^d)W_d(t) \quad (6.4)$$

is regular at $t = \rho_d$ and

$$L_d(\rho_d) = r_d^{d^2} H_d(r_d) > 0. \quad (6.5)$$

Proof. For $j < d$, the denominator $j+1 - r_d^j$ in (6.2) is positive. Induction gives $h_j(r_d) > 0$, so $H_d(r_d) > 0$. It follows that $h_d(q)$ has a simple pole at $q = r_d$, of residue $-H_d(r_d)/(d r_d^{d-1})$.

On the other hand, (4.8) and theorem 6.1 show that C_d has at most a simple pole there: only the denominator factor $q^d - (d+1)$ vanishes, and its derivative is $d r_d^{d-1} \neq 0$.

The singular parts of the terms $C_d(q)e^{q^d z}$ and $h_d(q)e^{(d+1)z}$ must cancel. Here is a justification that also allows collisions among other frequencies. Every Taylor coefficient of f_d in z is a polynomial in q , so the coefficient of $(q - r_d)^{-1}$ in its finite exponential representation is identically zero. For any other frequency, that Laurent coefficient is a polynomial in z times $e^{\lambda(r_d)z}$, obtained by finitely many Taylor terms of the exponential. All these exponents satisfy $\lambda(r_d) < d+1$. Polynomials times exponentials at distinct exponents are linearly independent: applying sufficiently high powers of $\partial_z - \lambda(r_d)$ annihilates each unwanted exponent and leaves a nonzero multiple of the remaining one. Therefore the coefficient of $e^{(d+1)z}$ must vanish separately. Since the two relevant amplitudes have at most simple poles, their residues sum to zero. This proves (6.3).

Finally, use $q = t^{-1}$ and $q^d - (d+1) = t^{-d}(1 - (d+1)t^d)$ in (4.2). The finite limiting value is exactly (6.5). $\square$

Theorem 6.3 (Exact exponential growth). For every $d \geq 1$, the radius of convergence of $W_d(t)$ is exactly $\rho_d = (d+1)^{-1/d}$, and

$$\limsup_{k \rightarrow \infty} a_{d,k}^{1/k} = (d+1)^{1/d}. \quad (6.6)$$

Proof. By theorems 4.2 and 6.1, there are no poles in $|t| < \rho_d$. By theorem 6.2, $t = \rho_d$ is an actual pole. The radius is therefore exactly ρ_d . The coefficient formula for the radius of convergence gives (6.6). $\square$

6.2 The periodic leading term

The same argument provides more than a limsup. Let $\zeta_j = e^{2\pi i j/d}$ and $u_j = \rho_d \zeta_j$. The function L_d in (6.4) is regular at all u_j : theorem 6.1 leaves at most the single simple factor $1 - (d+1)t^d$ on this circle.

Corollary 6.4 (Residue-class asymptotics). There is a number $R > \rho_d$ such that

$$a_{d,k} = \rho_d^{-k} \mathcal{A}_{d,k \bmod d} + O(R^{-k}), \quad (6.7)$$

where

$$\mathcal{A}_{d,r} = \frac{1}{d} \sum_{j=0}^{d-1} L_d(u_j) \zeta_j^{-r} \quad (0 \leq r < d). \quad (6.8)$$

Every $\mathcal{A}_{d,r}$ is real and nonnegative, and $\sum_r \mathcal{A}_{d,r} = L_d(\rho_d) > 0$.

Proof. At $t = u_j$, the contribution of the simple pole is

$$\frac{L_d(u_j)}{d} \frac{1}{1 - t/u_j}.$$

Subtracting these contributions removes all singularities on the first pole circle. Since the function is rational, the remainder is analytic in a strictly larger disk. A coefficient bound on an intermediate circle gives the error term in (6.7). Extracting the simple-pole coefficients yields (6.8). Along each residue class, $\rho_d^k a_{d,k}$ tends to $\mathcal{A}_{d,r}$; nonnegativity of the coefficients makes each limit nonnegative. Summing the discrete Fourier formula over r leaves only the term $j = 0$. $\square$

The theorem deliberately states a limsup for general d . Strict positivity of every residue-class amplitude has not been proved here. Nor is convergence of the consecutive-term ratio asserted; the explicit $d = 2$ formula shows that such a ratio limit can fail.

7 The OEIS partition connection: a structural contrast

Let $T(n, d)$ denote the A256193 partition triangle [4]. The convention matters: after writing the parts of an ordinary partition in nonincreasing order, choose exactly d positions to mark. Different markings of equal-part positions are distinct. Equivalently,

$$T(n, d) = \sum_{\lambda \vdash n} \binom{\ell(\lambda)}{d}, \quad (7.1)$$

where $\ell(\lambda)$ is the number of parts. Its two-variable generating function is

$$\sum_{n, d \geq 0} T(n, d) x^n y^d = \prod_{m \geq 1} \frac{1}{1 - (1 + y)x^m}. \quad (7.2)$$

Indeed, a block of r equal parts of size m has marking polynomial $(1 + y)^r$, and summing over r gives the m th geometric factor.

The initial diagonal agreement with W_d does not extend to the complete series. For example,

k	0	1	2	3	4	5	6	7
$a_{2,k}$	1	4	11	31	65	157	298	661
$T(k + 2, 2)$	1	4	11	24	49	89	158	262

The following result explains a global analytic distinction, without requiring a combinatorial interpretation of each discrepancy.

Theorem 7.1 (The partition diagonals are nonrational). *For every fixed $d \geq 1$, put*

$$V_d(t) = \sum_{k \geq 0} T(d + k, d) t^k. \quad (7.3)$$

Then V_d has radius of convergence 1 and is not rational. Moreover,

$$\limsup_{k \rightarrow \infty} \left(\frac{a_{d,k}}{T(d + k, d)} \right)^{1/k} = (d + 1)^{1/d} > 1. \quad (7.4)$$

Proof. There are elementary bounds

$$p(k) \leq T(d+k, d) \leq (d+k)^d p(d+k). \quad (7.5)$$

For the lower bound, append d marked singleton parts to any unmarked partition of k , placing them after the unmarked singleton parts. This is injective. For the upper bound, each partition of $d+k$ has at most $d+k$ positions available for marking.

For completeness, $p(n)$ is subexponential without needing a sharp partition asymptotic. For any fixed $0 < x < 1$,

$$\sum_{n \geq 0} p(n)x^n = \prod_{m \geq 1} (1 - x^m)^{-1} < \infty,$$

since

$$\sum_{m \geq 1} -\log(1 - x^m) \leq \sum_{m \geq 1} \frac{x^m}{1 - x^m} \leq \frac{x}{(1 - x)^2}.$$

Thus $\limsup p(n)^{1/n} \leq x^{-1}$; letting $x \uparrow 1$ and using $p(n) \geq 1$ gives $p(n)^{1/n} \rightarrow 1$. By (7.5), $T(d+k, d)^{1/k} \rightarrow 1$, proving the radius assertion and, together with theorem 6.3, (7.4).

It remains to prove nonrationality. Partition numbers cannot have a polynomial upper bound. If $p(n) = O(n^B)$, choose an integer $s > B + 1$. Taking each multiplicity of parts $1, \dots, s$ independently from $0, \dots, M$ produces $(M+1)^s$ different partitions, each of total size at most $N = Ms(s+1)/2$. But the assumed bound would imply

$$(M+1)^s \leq \sum_{n=0}^N p(n) = O(N^{B+1}) = O(M^{B+1}),$$

a contradiction as $M \rightarrow \infty$. By (7.5), the coefficients of V_d likewise admit no polynomial upper bound. A rational series of radius 1 does have such a bound: partial fractions express its coefficients as a finite sum of polynomial factors times exponential factors of modulus at most 1. Therefore V_d cannot be rational. $\square$

In particular, $W_d - V_d$ is nonrational and is not a polynomial. The difference cannot be repaired by changing finitely many coefficients. The earlier near-diagonal agreement concerns a finite initial region for fixed d ; the full tails belong to different generating-function classes and have different exponential scales. This conclusion does not claim an eventual coefficientwise inequality for every k : (7.4) is the precise statement proved.

8 Algorithms and independent exact verification

8.1 Constructing the rational functions

The program `code/derive.py` implements (3.4)–(3.9). A spectrum is a dictionary whose key is the coefficient vector $(c_0, c_1, \dots)$ of a frequency and whose value is an exact rational function in q . Multiplication adds keys and multiplies amplitudes; equal frequencies are combined before solving the next differential equation.

At each level the program verifies, in $\mathbb{Q}(q)$, both the differential identity and the initial condition:

$$(\lambda - q^d)c_{d,\lambda} = h_{d,\lambda}, \quad \sum_{\lambda} c_{d,\lambda} = 0.$$

It extracts C_d , reverses its numerator and denominator directly, checks coprimality, and verifies that the resulting reduced denominator divides the universal denominator. Direct coefficient reversal avoids the expression swell of substituting $q = 1/t$ into a large expanded rational expression.

The finite frequency count M_d is not by itself a bit-complexity bound. Rational-function numerator and denominator sizes also grow, and cancellation can be expensive. No polynomial-time claim in d is made. For a fixed derived denominator of degree L , however, successive tail coefficients take $O(L)$ integer arithmetic operations each. Producing K coefficients takes $O(KL)$ such operations, apart from integer bit costs. Standard polynomial reduction for a linear recurrence also permits an isolated distant term to be computed in $O(L^2 \log K)$ arithmetic operations using naive polynomial multiplication.

8.2 An independent path from the original recurrence

The verifier code/`verify.py` uses only Python’s standard library. It does not call SymPy and does not reconstruct the finite spectra. Instead it evaluates the original recurrence after reversal, truncating in deficit degree. The degree gaps in section 2 show that all shifts are nonnegative, so discarding higher deficit coefficients cannot affect lower ones.

For every $1 \leq d \leq 6$ and $0 \leq k \leq 40$, it compares the rational-series coefficient against the original recurrence at three lengths:

$$n = d + k + 1, \quad n = d + k + 2, \quad n = 48.$$

These are $6 \cdot 41 \cdot 3 = 738$ exact integer comparisons. The first length is the proved stabilization boundary; the other two check that the same coefficient persists. In addition, all 251 exported coefficients of W_1 and W_2 are checked against their independent closed formulas. No floating-point arithmetic is involved.

Table 2: A prefix recomputed by both methods. Entries are $a_{d,k}$.

k	$d = 1$	$d = 2$	$d = 3$	$d = 4$	$d = 5$	$d = 6$
0	1	1	1	1	1	1
1	3	4	5	6	7	8
2	7	11	16	22	29	37
3	15	31	41	63	92	129
4	31	65	112	155	247	376
5	63	157	244	393	590	966
6	127	298	542	869	1404	2258
7	255	661	1155	1863	3079	5140
8	511	1196	2288	3886	6471	11078
9	1023	2516	4552	7863	13216	22921
10	2047	4434	8852	15217	26354	46011
11	4095	9002	16403	29134	50977	90269

The symbolic formulas prove statements for all indices. The finite comparisons serve a different purpose: they guard against transcription, normalization, implementation, and indexing errors. The computations are not used as a substitute for the general arguments.

8.3 Reproduction and files

From the archive root, the following commands reproduce the exact results:

```
python -m pip install -r requirements.txt
python code/derive.py --max-d 6 --check-only
python code/verify.py --max-k 40 --export-count 251
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The derivation was run with SymPy 1.14.0. The independent verifier needs Python 3.9 or later and no third-party package. The data directory contains normalized numerator and denominator arrays in ascending degree order, exact spectral certificates, CSV tables, individual coefficient files, and the actual verification logs. No Lean or other proof-assistant formalization is claimed.

9 What is settled, and what remains

The whole-series rationality problem has a uniform affirmative answer, with explicit finite algebraic data. The proof depends only on the exact recurrence (1.1). It establishes an integer recurrence for every descent count, an exact radius of convergence, and periodic residue-class asymptotics. The two-descent case is completely explicit in elementary sequences.

Several sharper questions remain beyond this report. A general rule for the cancellations in $\mathcal{D}_d$ would determine minimal recurrence orders without symbolic gcd computations. A proof that every $\mathcal{A}_{d,r}$ is strictly positive would upgrade the general limsup to an ordinary k th-root limit. A direct combinatorial explanation of the rational numerators or the complete discrepancy $a_{d,k} - T(d+k, d)$ could make the algebra more transparent. Finally, the growth and structure of the bivariate series $\sum_{d \geq 1} u^d W_d(t)$ is a separate problem; fixed- d rationality does not imply its rationality in two variables.

The useful outcome of this investigation is therefore not a fitted recurrence or a large computational prefix. It is a finite-spectrum mechanism that determines each complete stabilized generating function, together with a precise distinction between those series and the OEIS partition diagonals that originally suggested them.

A The spectral construction in pseudocode

In the following pseudocode, dictionary keys are polynomials in q and values lie in $\mathbb{Q}(q)$. Zero entries are removed after every collection. The single-symbol key q^d refers to a frequency, not to its amplitude.

```

F[0] := { 1 : 1 }
for d := 1, ..., D:
    G := empty dictionary
    for a := 0, ..., d-1:
        b := d-1-a
        for (lambda, c) in F[a]:
            for (mu, e) in F[b]:
                G[lambda+mu] += q^b * c * e
    for (lambda, c) in F[d-1]:
        G[lambda] -= q^(d-1) * c

    F[d] := empty dictionary
    for (lambda, h) in G:
        F[d][lambda] := h / (lambda - q^d)
    F[d][q^d] := -sum(values of F[d])

    C[d](q) := F[d][q^d]
    W[d](t) := t^(-d*(d+1)) * C[d](1/t)
    cancel and normalize the denominator to have constant term 1

```

The decisive invariant is $\deg \lambda < d$ for every key in G . It guarantees that the division is legal over $\mathbb{Q}(q)$ and that the q^d mode is the unique mode surviving the formal leading-tail limit.

B A compact audit of claims

Claim	Basis and qualification
Definition of the polynomial family	The published recurrence [2, Theorem 4.1], reproduced as (1.1).
Existence of W_d	Known stabilization result [2]; independently proved from the recurrence in theorem 2.2.
Rationality for every d	General proof in theorems 3.2, 4.1 and 4.2; not inferred from experiments.
Exact growth radius	Pole exclusion and a noncancellation proof in theorems 6.1 to 6.3.
Closed formulas for $d = 2$	Exact differential calculation, rational identity, and coefficient extraction in section 5.
Denominator degrees through $d = 6$	Exact rational-function arithmetic, coprimality checks, and retained machine-readable results.
Coefficient verification	738 comparisons with the original recurrence, plus closed-form checks; all passed. This is an implementation audit, not proof-assistant verification.
Novelty	No all- d rationality theorem was found in the three primary papers reviewed. No claim is made that the entire literature has been exhaustively checked.

References

- [1] W. Dugan, S. Glennon, P. E. Gunnells, and E. Steingrímsson, *Tiered trees, weights, and q -Eulerian numbers*, Journal of Combinatorial Theory, Series A **164** (2019), 24–49. arXiv:1702.02446. The motivating discussion of stabilized series and partition diagonals is Section 6.12.
- [2] A. Agrawal, C. Choi, and N. Sun, *On Permutation Weights and q -Eulerian Polynomials*, Annals of Combinatorics **24** (2020), 363–378. doi:10.1007/s00026-020-00493-5. arXiv:1809.07398, version 4.
- [3] E. Huang and R. Tang, *Minimum Decomposition on Maxmin Trees*, arXiv:2310.14385 (2023), version 2. The end of Section 5 discusses the challenge of reaching coefficients beyond the near-diagonal partition correspondence.
- [4] The OEIS Foundation Inc., A256193: Number $T(n, k)$ of partitions of n into two sorts of parts having exactly k parts of the second sort, The On-Line Encyclopedia of Integer Sequences. The marking convention is made explicit in (7.1).

Primary sources and the OEIS entry were consulted on September 19, 2026. All displayed new formulas and computational tables were derived or recomputed for this report.